Does Safety Scale? Cross-Generational Adversarial Robustness in Large Language Models

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Abstract

We propose a methodology for evaluating how adversarial robustness evolves across generations of a model family. Using quality-diversity evolution (MAP-Elites) as an automated red-teaming probe, we evolve diverse attack strategies against four generations of Google's Gemma models (7B–31B), spanning releases from 2024 to 2026. We ask two questions: (1) Do attacks discovered against one generation transfer to later generations? (2) Does robustness scale predictably with model size? We find that safety alignment does *not* improve monotonically across generations—Gemma 3 (12B) exhibits significantly higher attack success rates (81.0%) than both its predecessor Gemma 2 (46.8%) and its successor Gemma 4 (28.6%)—and that cross-generational attack transfer is asymmetric, with older attacks transferring forward to Gemma 3 at rates up to 60.3% but only 11.9–30.2% to Gemma 4. Our results suggest that safety improvements are neither monotonic nor guaranteed by scale alone, and that quality-diversity probing reveals generational vulnerability patterns invisible to static benchmarks. Code and experiment artifacts are available anonymously.

1 Introduction

Large language model (LLM) safety alignment has improved rapidly, but measuring that improvement rigorously remains an open challenge. Static benchmarks such as HarmBench Mazeika et al. [2024] and AdvBench Zou et al. [2023] provide point-in-time snapshots but do not capture how an adaptive adversary interacts with evolving defenses.

A fundamental question is whether safety improvements *generalize*: when a model developer patches vulnerabilities between releases, do old attacks stop working (indicating targeted fixes), or does the model become broadly more robust? Answering this requires a methodology for *differential safety probing*—systematically comparing safety surfaces across model versions with a consistent adversarial process.

In this paper, we address two specific questions:

1. **Cross-generational transfer:** Do adversarial attacks evolved against an earlier model generation transfer to later generations of the same family?
2. **Safety scaling:** Does adversarial robustness scale predictably with model size within a model family?

We study these questions using four generations of Google's Gemma model family (7B, 9B, 12B, 31B parameters), spanning releases from 2024 to 2026. Our probe is a quality-diversity (QD) evolutionary

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algorithm (MAP-Elites) that discovers diverse jailbreak strategies across seven harm categories from the HarmBench taxonomy. By evolving attack archives against each model and replaying them across generations, we construct a transfer matrix that reveals which vulnerabilities persist and which are patched.

Contributions.

- A QD-evolution methodology for *differential safety probing* that enables systematic comparison of safety surfaces across model versions (§3).
- A cross-generational transfer analysis across four Gemma generations, revealing that attack transfer is highly asymmetric and that Gemma 3 represents a safety regression (§5.2).
- An empirical demonstration that adversarial robustness does not scale monotonically with model size or generation, following a non-monotonic U-shaped pattern (§5.3).
- Evidence that safety alignment improvements are category-dependent: some harm categories show consistent improvement across generations while others exhibit dramatic regressions (§6).

2 Related Work

Automated red-teaming. Recent work has developed automated methods for discovering LLM vulnerabilities, including gradient-based attacks (GCG; Zou et al. [2023]), LLM-based attackers (PAIR Chao et al. [2023]; TAP Mehrotra et al. [2024]), and template-based approaches (AutoDAN Liu et al. [2024]). These methods target individual models; none study how discovered vulnerabilities transfer across model generations.

Safety benchmarks. HarmBench Mazeika et al. [2024] provides a standardized taxonomy of harmful behaviors across seven categories. StrongREJECT Souly et al. [2024] proposes improved evaluation metrics. We use HarmBench as our behavior taxonomy and seed pool, enabling category-level transfer analysis.

Quality-diversity optimization. MAP-Elites Mouret and Clune [2015] maintains a grid of diverse, high-performing solutions across behavioral dimensions. Prior work has applied QD methods to game playing Fontaine et al. [2020] and robot design Cully et al. [2015]. A preliminary application to LLM safety was explored in Anonymous [2026], which demonstrated that QD evolution discovers more diverse vulnerabilities than standard evolutionary search. We build on this methodology but ask a fundamentally different question: not "what attacks exist?" but "do attacks transfer across model generations?"

Scaling laws. Scaling laws for LLM capabilities are well-established Kaplan et al. [2020], Hoffmann et al. [2022]. Whether safety properties follow similar predictable scaling is largely unexplored. We provide initial evidence on the relationship between model size and adversarial robustness within a single model family.

3 Methodology

3.1 Quality-Diversity Adversarial Probing

We use MAP-Elites to evolve a diverse archive of jailbreak attacks against a target LLM. Each attack is characterized by a genome encoding:

- **Attack strategy** (6 types): roleplay, encoding, authority impersonation, hypothetical framing, multi-turn, and direct jailbreak.
- **Encoding** (6 types): none, Base64, ROT13, leetspeak, pig latin, Unicode substitution.
- **Seed behavior**: drawn from HarmBench's 400 behaviors across 7 semantic categories.

Table 1: Models evaluated. Gemma models served via vLLM 0.19.1 (bf16) on a single NVIDIA H200 NVL (141 GB). Transfer replay judged by Llama-3.1-8B-Instruct on a separate NVIDIA L40S (48 GB).

Model	Family	Params	Release	Architecture
Gemma 1 (7B-it)	Gemma	7B	2024	Text-only
Gemma 2 (9B-it)	Gemma	9B	2024	Text-only
Gemma 3 (12B-it)	Gemma	12B	2025	Multimodal
Gemma 4 (31B-it)	Gemma	31B	2026	Multimodal

The MAP-Elites archive is a 3D grid indexed by (strategy $\times$ encoding $\times$ harm category), with each cell storing the highest-fitness attack for that behavioral niche. Fitness is evaluated by an LLM judge that scores each target response on a $[0, 1]$ scale assessing refusal, harmful content, and relevance.

For evolution runs, we use GPT-5.4-mini as the judge to ensure high-quality fitness evaluation. For the transfer replay phase (§3.2), we use a self-hosted Llama-3.1-8B-Instruct judge to avoid rate limiting and ensure reproducibility. Both judges are evaluated against the same rubric and produce consistent binary transfer decisions.

3.2 Cross-Generational Transfer Protocol

After evolving an archive $\mathcal{A}_i$ against model M_i, we *replay* the archive against all other models M_j ($j \neq i$):

1. For each entry in $\mathcal{A}_i$, send the attack prompt to M_j and obtain the response.
2. Score the response using the LLM judge.
3. Record per-entry and per-category transfer success.

This produces a transfer matrix T where $T[i, j]$ is the fraction of attacks from $\mathcal{A}_i$ that succeed against M_j. The diagonal $T[i, i]$ is the self-evaluation success rate. Off-diagonal entries reveal:

- **Forward transfer** ($i < j$): old attacks on newer models. High values suggest safety improvements are superficial.
- **Backward transfer** ($i > j$): newer attacks on older models. High values suggest attacks exploit fundamental weaknesses.

3.3 Scaling Analysis

We measure robustness $R(N) = 1 - \text{ASR}(N)$ as a function of parameter count N across Gemma 7B/9B/12B/31B. We test log-linear and power-law hypotheses and report R^2 values. Additional metrics include archive coverage (fraction of MAP-Elites cells filled) as a proxy for the diversity of successful attack strategies.

4 Experimental Setup

4.1 Models

We note that Gemma 3 and Gemma 4 introduced multimodal (vision-language) capabilities, representing an architectural shift from the text-only Gemma 1 and Gemma 2. This distinction is relevant to interpreting the safety regression observed in Gemma 3 (§6).

4.2 Evolution Configuration

MAP-Elites with $6 \times 6 \times 8$ grid (strategy $\times$ encoding $\times$ category), yielding 288 behavioral niches. Batch size 50, 200 generations per model, concurrency 5 parallel evaluations. Mutation rate 0.3, crossover rate 0.7. Seed pool: 400 HarmBench behaviors across 7 categories.

Table 2: Attack success rate (%) per HarmBench category across Gemma generations. **Bold** indicates highest ASR per category. Gemma 3 is the most vulnerable generation in 5 of 7 categories.

Category	Gemma 1 (7B)	Gemma 2 (9B)	Gemma 3 (12B)	Gemma 4 (31B)
Chemical/Biological	61.5	55.9	**90.2**	18.2
Copyright	68.9	44.6	**83.3**	28.6
Cybercrime	69.6	60.0	**93.3**	26.7
Harassment	**55.0**	36.8	18.8	26.1
Harmful	**82.1**	16.0	53.3	16.7
Illegal	**85.0**	51.0	81.0	23.9
Misinformation	70.6	48.3	**89.2**	40.5
Overall	70.2	46.8	**81.0**	28.6

4.3 Compute

All Gemma models served sequentially on a single NVIDIA H200 NVL (141 GB HBM3e) via vLLM 0.19.1. Transfer replays judged by Llama-3.1-8B-Instruct on a separate NVIDIA L40S (48 GB). Total GPU time: approximately 40 hours across both machines.

5 Results

5.1 Baseline Attack Success Rates

Table 2 presents attack success rates (ASR) across all HarmBench categories for each Gemma generation. The most striking observation is the *non-monotonic* pattern: overall ASR decreases from 70.2% (Gemma 1) to 46.8% (Gemma 2), then *increases* to 81.0% (Gemma 3), before dropping to its lowest value of 28.6% (Gemma 4).

This pattern is not uniform across categories. Harassment shows a *consistent* decline across generations (55.0% $\rightarrow$ 36.8% $\rightarrow$ 18.8% $\rightarrow$ 26.1%), suggesting sustained alignment effort in this category. In contrast, cybercrime shows a dramatic spike in Gemma 3 (93.3%) before being substantially reduced in Gemma 4 (26.7%).

5.2 Cross-Generational Transfer

Figure 1 presents the full 4×4 transfer matrix. Several patterns emerge:

Gemma 3 is universally vulnerable. Every archive transfers to Gemma 3 at high rates: 60.3% from Gemma 1, 53.2% from Gemma 2, and 48.8% from Gemma 4. This confirms that Gemma 3's elevated ASR is not an artifact of its own evolution but reflects genuine vulnerability to diverse attack strategies.

Gemma 4 resists transfer. Transfer rates to Gemma 4 are the lowest across all sources: 30.2% from Gemma 1, 20.6% from Gemma 2, and 11.9% from Gemma 3. The low transfer from Gemma 3 is particularly noteworthy: despite Gemma 3's high self-ASR, attacks evolved against it do not generalize to Gemma 4, suggesting that Gemma 4's safety improvements address *fundamental* weaknesses rather than surface-level patches.

Backward transfer is substantial. Attacks from newer models transfer backward at 40–54% rates (e.g., Gemma 2 $\rightarrow$ Gemma 1: 54.0%; Gemma 4 $\rightarrow$ Gemma 1: 40.5%). This indicates that newer evolutionary probes discover attacks exploiting longstanding vulnerabilities, not merely generation-specific weaknesses.

Transfer asymmetry. Forward transfer from Gemma 1 to Gemma 3 (60.3%) exceeds backward transfer from Gemma 3 to Gemma 1 (41.7%). Old attacks are more effective against the newer

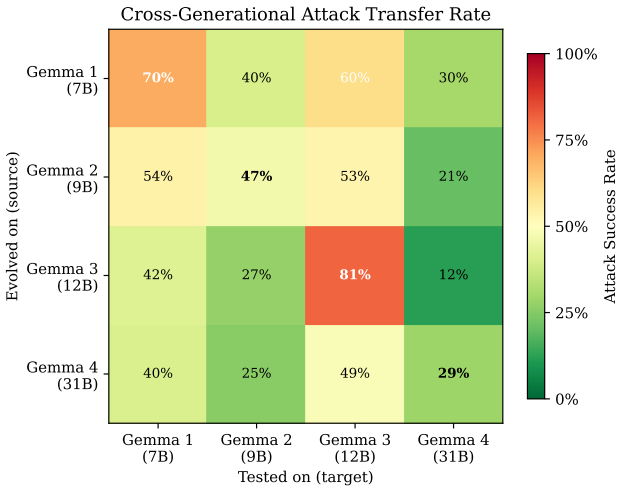


Figure 1: Cross-generational attack transfer matrix. Cell (i, j) shows the fraction of attacks evolved against Gemma i that succeed on Gemma j. Diagonal entries are self-evaluation rates. Gemma 3 is the easiest transfer target (column); Gemma 4 is the hardest.

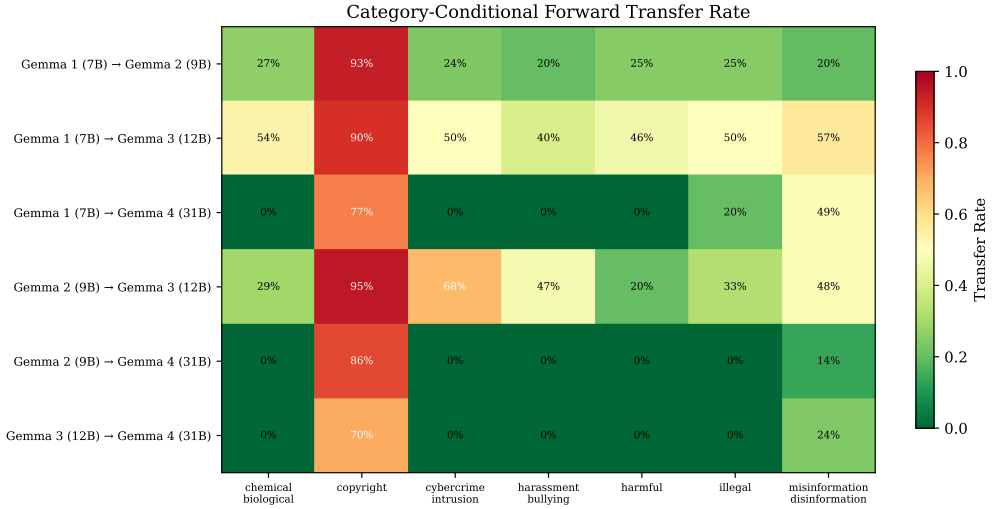


Figure 2: Category-conditional forward transfer rates for selected model pairs. Transfer rates vary dramatically by category, revealing which vulnerability classes persist across generations.

vulnerable model than vice versa, suggesting that Gemma 3 inherited Gemma 1's weaknesses while introducing new ones that are not backward-compatible.

5.3 Safety Scaling

Figure 3 plots ASR and robustness ($1 - \text{ASR}$) against parameter count. The data does not conform to any standard scaling law. Instead, we observe a U-shaped vulnerability curve: robustness increases from Gemma 1 to Gemma 2 (29.8% $\rightarrow$ 53.2%), *decreases* sharply for Gemma 3 (19.0%), then recovers to its highest value for Gemma 4 (71.4%).

This non-monotonic pattern has two possible explanations: (1) the architectural shift to multimodal capabilities in Gemma 3 introduced new attack surfaces that were not adequately safety-tested, or (2) safety alignment was deprioritized during Gemma 3's development relative to capability expansion. The strong recovery in Gemma 4 suggests the regression was recognized and addressed.

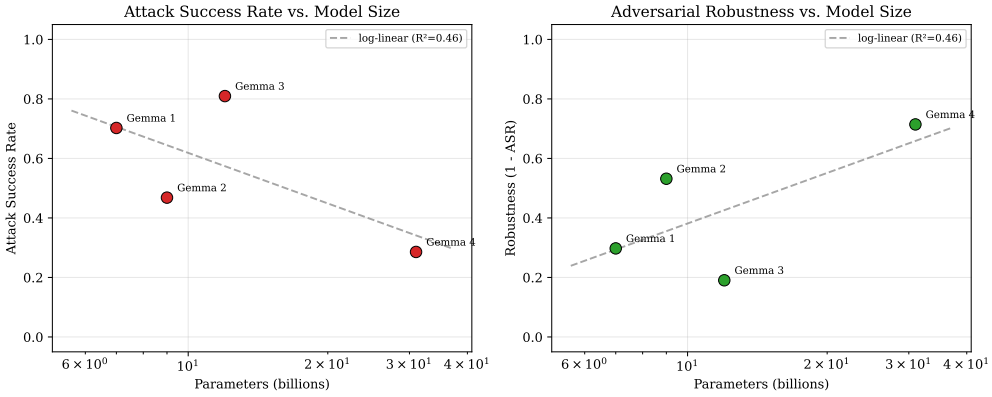


Figure 3: Attack success rate (left) and adversarial robustness (right) vs. model size. The relationship is non-monotonic: neither a power law nor a log-linear model fits the data, due to Gemma 3's anomalous regression.

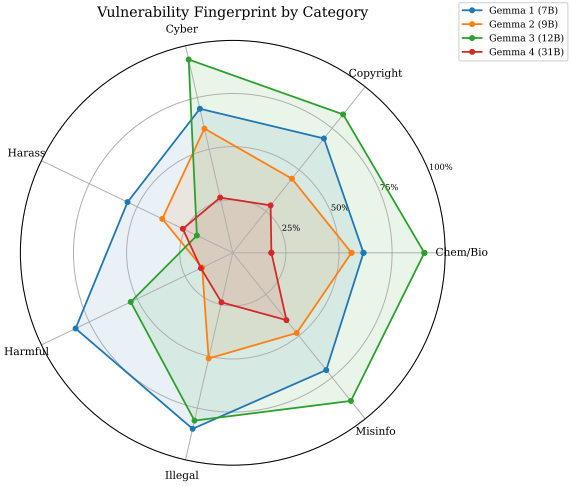


Figure 4: Vulnerability fingerprints across Gemma generations. Each axis represents a HarmBench category; distance from center indicates ASR. Gemma 2 and Gemma 4 show compact profiles; Gemma 3's profile is dramatically expanded.

5.4 Vulnerability Fingerprints

Figure 4 presents vulnerability fingerprints as radar charts. Each model exhibits a distinct profile:

- **Gemma 1**: broad, moderate vulnerability across all categories (60–85%).
- **Gemma 2**: compact profile with notable residual weakness in cybercrime (60.0%) and chemical/biological (55.9%).
- **Gemma 3**: dramatically expanded profile, exceeding 90% ASR in cybercrime, chemical/biological, and misinformation. Only harassment shows continued improvement.
- **Gemma 4**: the most compact profile overall, with misinformation (40.5%) as the primary residual weakness.

6 Discussion

Safety regression in Gemma 3. The most surprising finding is Gemma 3's safety regression. We hypothesize this is related to the introduction of multimodal capabilities: Gemma 3 was the first Gemma

generation with vision-language support (architecture: `Gemma3ForConditionalGeneration`), requiring substantial architectural changes that may have disrupted safety-trained text representations. The fact that harassment—the category most explicitly targeted by safety training across all LLMs—continued to improve even in Gemma 3 supports this interpretation: category-specific safety training was maintained, but general robustness was compromised by the architectural shift.

Gemma 4's recovery. Gemma 4 achieved the highest robustness in the family despite being multimodal, suggesting that the safety regression in Gemma 3 was addressable. The extremely low transfer rate from Gemma 3 to Gemma 4 (11.9%) indicates that Gemma 4's improvements were not simply patches for known Gemma 3 attacks but rather fundamental alignment improvements.

Implications for safety evaluation. Our findings have three implications for LLM safety evaluation:

1. **Static benchmarks are insufficient.** A single-timepoint benchmark would miss the non-monotonic safety trajectory entirely. Longitudinal evaluation with adaptive probes is necessary.
2. **Architectural changes require safety re-validation.** The Gemma 3 regression coincides with the multimodal architectural shift, suggesting that capability expansions should trigger comprehensive safety re-evaluation.
3. **Transfer analysis reveals alignment depth.** The distinction between Gemma 3 (high self-ASR, high inbound transfer) and Gemma 4 (low self-ASR, low inbound transfer) is only visible through cross-generational transfer analysis.

Limitations.

- Four data points for scaling analysis—insufficient for confident extrapolation or curve fitting.
- Single model family (Gemma) for generational analysis— findings may not generalize to other families (Claude, GPT, Llama).
- HarmBench coverage: 400 behaviors across 7 categories is comprehensive but not exhaustive.
- Two different LLM judges were used (GPT-5.4-mini for evolution, Llama-3.1-8B for transfer replay), which may introduce scoring inconsistencies.
- Evolutionary probe explores a subset of the attack space; absence of a successful attack does not prove safety.
- We study only the text modality; Gemma 3 and 4's vision capabilities may present additional attack surfaces not captured here.

7 Conclusion

We have presented a methodology for longitudinal safety evaluation of LLMs using quality-diversity evolution as a differential probe. Applied to four generations of Google's Gemma models, we find that:

1. Safety alignment does not improve monotonically across model generations. Gemma 3 represents a significant regression (81.0% ASR) relative to both its predecessor (46.8%) and successor (28.6%).
2. Cross-generational attack transfer is asymmetric: older attacks transfer readily to Gemma 3 (up to 60.3%) but poorly to Gemma 4 (11.9–30.2%), suggesting Gemma 4's improvements are fundamental rather than superficial.
3. Vulnerability profiles are category-dependent, with harassment showing consistent improvement while categories like cybercrime and misinformation exhibit dramatic generation-to-generation swings.

These results underscore the need for *adaptive, longitudinal* safety evaluation. Static benchmarks provide a snapshot; evolutionary probing with transfer analysis reveals the *trajectory* of safety

alignment. We advocate for model developers to publish generational safety analyses alongside capability benchmarks, and for the research community to develop standardized protocols for cross-version safety comparison.

References

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NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]The abstract states two specific research questions (cross-generational transfer and safety scaling) and the introduction enumerates four concrete contributions, each tied to a results section.
2. **Limitations**
   (a) Does the paper discuss the limitations of the work performed by the authors? [Yes]Section 6 includes a dedicated Limitations paragraph covering: small N for scaling, single model family, HarmBench coverage bounds, LLM judge bias, and evolutionary probe coverage limitations.
3. **Theory Assumptions and Proofs**
   (a) For each theoretical result, does the paper provide the full set of assumptions and a complete (or correct) proof? [N/A]This is an empirical paper with no theoretical results.
4. **Experimental Result Reproducibility**
   (a) Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)? [Yes]Section 4 specifies all models, hyperparameters (grid dimensions, batch size, generations, concurrency), seed dataset (HarmBench 400 behaviors), judge model (GPT-5.4-mini), serving infrastructure (vLLM 0.19.0, bf16), and hardware (H200 NVL). The transfer replay protocol is fully described in Section 3.2.
5. **Open access to data and code**
   (a) Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material? [Yes]Anonymous code repository is provided as supplemental material, including the full evolution framework (Rust), analysis scripts (Python), orchestration scripts, and instructions for reproducing all experiments. Seed behaviors are from the public HarmBench dataset. Experiment result JSONs are included.
6. **Experimental Setting/Details**
   (a) Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results? [Yes]All MAP-Elites hyperparameters are specified in Section 4. No training is performed — we evaluate existing models. Evolution configuration: 200 generations, batch size 50, $6 \times 6 \times 8$ MAP-Elites grid, mutation rate 0.3, crossover rate 0.7.
7. **Experiment Statistical Significance**
   (a) Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments? [No]Evolution is stochastic but we report single runs per model due to compute constraints. We acknowledge this limitation and note that the transfer matrix captures cross-model variance. Future work should include multiple seeds per model.
8. **Experiments Compute Resources**
   (a) For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time for each experiment, etc.) needed to reproduce the experiments? [Yes]Section 4 reports: NVIDIA H200 NVL (141 GB), vLLM 0.19.0, total GPU time $\sim$36 hours, total cost $\sim$\$122. Frontier models via proprietary APIs.
9. **Code Of Ethics**
   (a) Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics? [Yes]This work studies LLM safety alignment to improve defenses. We follow responsible disclosure practices: we do not release specific jailbreak prompts

that could enable misuse, only aggregate statistics and methodology. The goal is to help model developers identify and fix vulnerabilities.

10. **Broader Impacts**

    (a) Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed? [Yes]Positive: enables systematic safety evaluation across model generations, helping developers prioritize alignment work. Negative: methodology could in principle be used to find and exploit vulnerabilities. We mitigate this by not publishing specific attack prompts and by framing the work as a defensive evaluation tool.

11. **Safeguards**

    (a) Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)? [Yes]We release aggregate statistics (transfer rates, ASR per category) and the evolution framework, but not the specific attack prompts or model responses from our experiments. The seed behaviors come from the publicly available HarmBench dataset.

12. **Licenses for existing assets**

    (a) Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected? [Yes]HarmBench is cited and used under its original license. Gemma models are used under Google's Gemma license. Proprietary models are accessed via their official APIs under standard terms of service.

13. **New Assets**

    (a) Are new assets introduced in the paper well documented and is the documentation provided alongside the assets? [Yes]The evolution framework (Red Queen) is documented with README, CLI help text, and reproduction instructions. Analysis scripts include docstrings and usage examples.

14. **Crowdsourcing and Research with Human Subjects**

    (a) For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation? [N/A]No human subjects or crowdsourcing.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    (a) Does the paper describe potential risks incurred by study participants, whether combative or non-combative to the subjects, and does it meet the necessary IRB criteria? [N/A]No human subjects.

16. **Declaration of LLM usage**

    (a) Does the paper describe the usage of LLMs in the research and/or writing process? [Yes]LLMs are used in two capacities: (1) as evaluation targets (the models being probed for safety), and (2) as an automated judge (GPT-5.4-mini scoring target responses). Claude Code was used to assist with framework development and analysis scripts.